BIOT6900 Module 4 Review

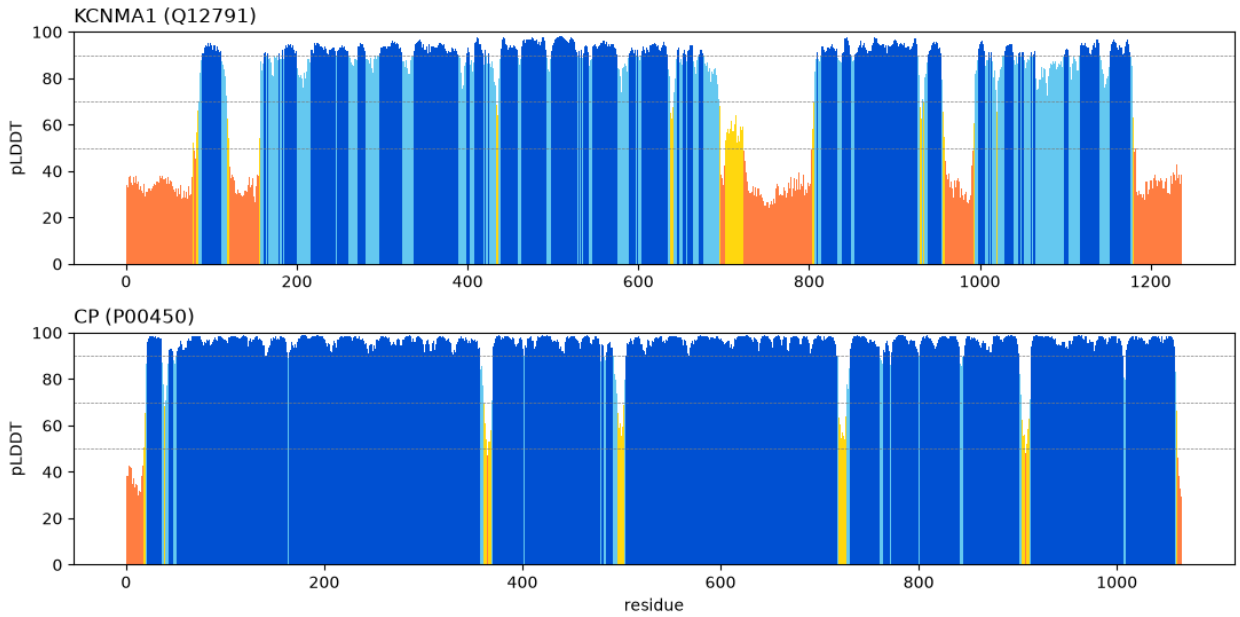
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Background

In Module 2, we identified various genes that were potentially relevant to a disease. I chose colon cancer, which is the third most common cancer diagnosed in both men and women in the United States and the second most common cause of cancer deaths.[1] There has been an interesting trend linking colon cancer to increasingly younger populations, which has intensified the search for new drugs to treat the disease. In Module 3, we further refined that list by introducing a tractability prior, which compared our selected genes to similar ones, and judged how druggable they are based on prior data. Now, integrating the output of modules 2 and 3, we select our top two candidates and predict their structure using AlphaFold, allowing us to make an assessment of how druggable they truly are, as well as identify potential areas within the protein that can be targeted. Based on the evidence from the previous modules, I chose to investigate the protein-coding genes KCNMA1 and CP. KCNMA1 codes for a calcium-activated potassium channel which exists in the cell membrane, and CP codes for ceruloplasmin, which is a secreted copper binding glycoprotein.

What can be trusted

First, I downloaded the AlphaFold models for KCNMA1 and CP, and read the pLDDT (predicted Local Distance Difference Test) for each protein. Each residue within the protein is given a confidence score from 0-100. Lower scores indicate that the model is unsure of the exact location, usually indicating a disordered region. Predictions from these areas should be viewed skeptically. Higher scores (90+) mean that AlphaFold is very confident about its prediction. Delineated by color, these are the pLDDT graphs for KCNMA1 and CP, with related stats:

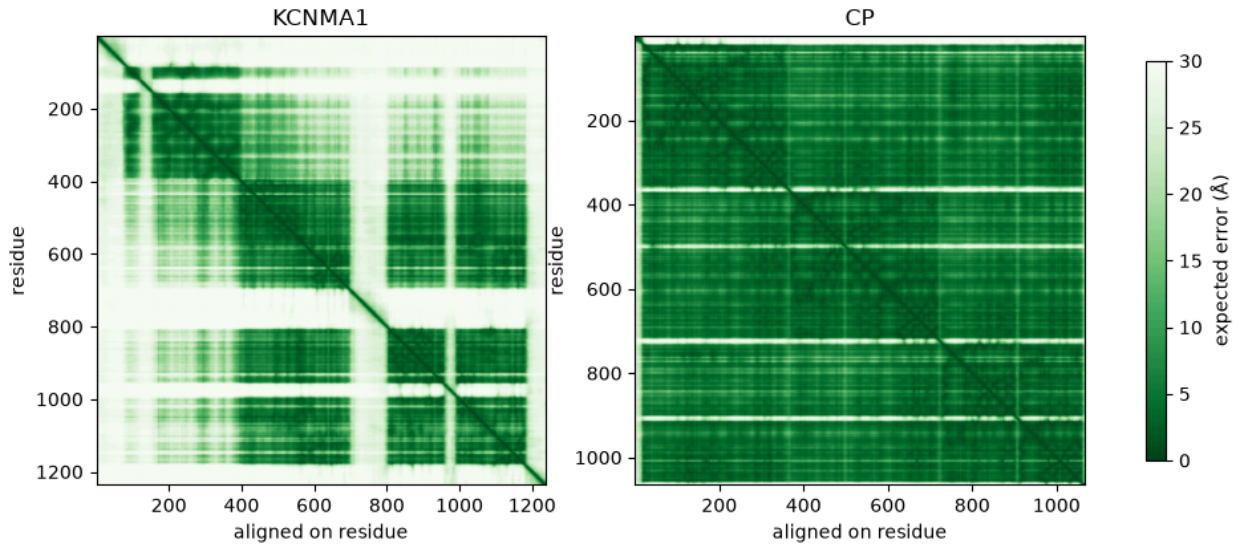


	gene	residues	mean_pLDDT	pct_above_90	pct_below_50	disordered_stretches
0	KCNMA1	1236	76.0	47	24	1-78, 121-155, 724-803, 959-992, 1181-1236
1	CP	1065	93.7	90	2	1-17

[Figure 1: pLDDT along the sequence for KCNMA1 and CP (insert from notebook, Step 3)]

First looking at KCNMA1, the mean_pLDDT is middling at 76.0, largely due to sizable sections of disorder. Nearly ¼ of the predictions are below 50% confidence. The predictions here closely match what is known of the protein's structure. Looking at the UniProt entry for KCNMA1, there is a known disordered stretch between positions 1-61, which lines up with AlphaFold's low confidence in that area. Other low confidence regions also align with other disordered regions. The experimentally derived crystal structure agrees about the disorder. 6V5A (UniProt numbering) leaves 679–802 and 957–991 unresolved. Those are the same areas AlphaFold marks below 50, so two independent methods find the same disorder. In contrast to KCNMA1, CP's pLDDT graph looks much better. The mean_pLDDT score is 93.7, meaning AlphaFold is highly confident in its structure prediction. There is only one small disordered stretch at the beginning in positions 1-17, which is the disordered signal peptide.[2] Otherwise, nearly the entire prediction can be trusted.

PAE graphs for reference:

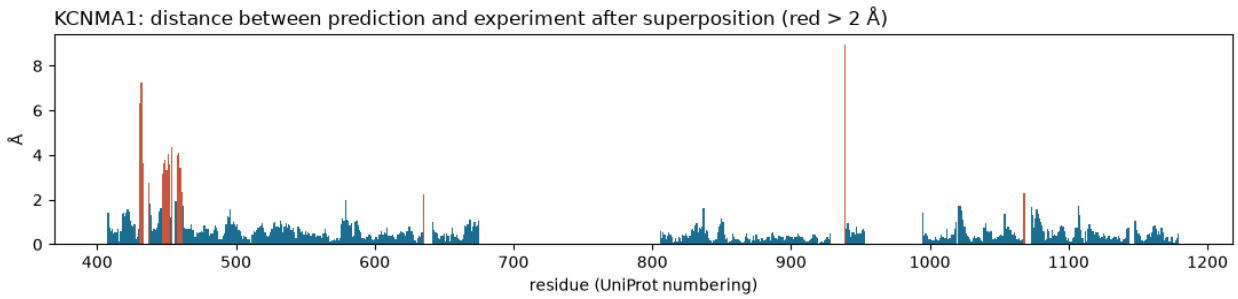


[Figure 2: PAE heatmaps for KCNMA1 and CP (insert from notebook, Step 4)]

The PAE (Predicted Aligned Error) graph answers a different question than pLDDT. pLDDT asks whether each piece is the right shape, while PAE asks whether the pieces are arranged correctly relative to each other. It should be noted that I modified the given code to implement a different PAE cutoff, reducing it from 8 Å to just 2 Å . With higher values for this cutoff, the different regions visible in the PAE graph for KCNMA1 were not being discriminated properly. With a value of 2 Å, the domains were correctly split in two. For KCNMA1, the domains were 85–398 and 399–1179, with a mean PAE of 22.1 Å between them. I would trust pocket analysis inside either KCNMA1 domain, but not any pocket that spans the two. CP had one domain, 20–1059.

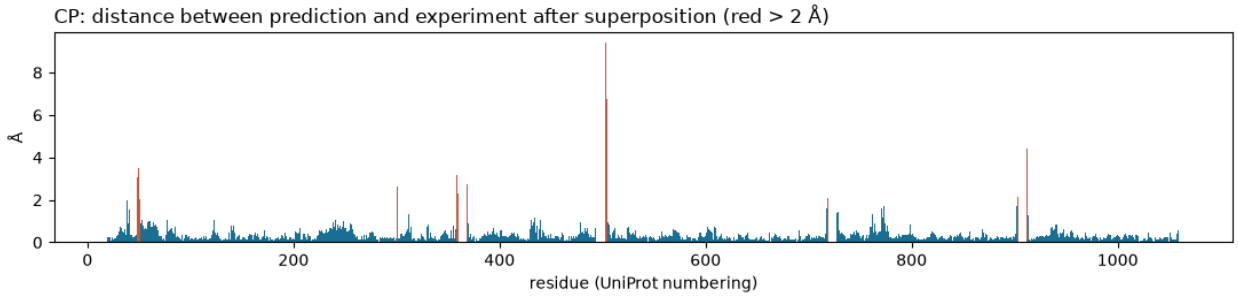
Prediction vs experiment

Both targets had experimental structures, which gives us a direct look at how far the predictions can be trusted. Starting with KCNMA1, the 6V5A structure I used for comparison is limited to only parts of the protein. The RMSD is 1.00 Å over 572 identical, confident residues. Looking into the one large region of experimental divergence, residues 451–459 sit around a mutation in the measured crystal structure. [3] This is something the crystal has that the prediction lacks, not a prediction error.



[Figure 3: KCNMA1 per-residue deviation from 6V5A after superposition (insert from notebook, Step 5)]

For CP, besides a few outliers, the prediction largely matches the experimental structure.[4] CP contains 6 Copper atoms which are not modeled by AlphaFold, the few large prediction errors are likely connected to those.



[Figure 4: CP per-residue deviation from 4ENZ after superposition (insert from notebook, Step 5)]

Druggability calls

KCNMA1: top five pockets by druggability score

pocket	Druggability Score	Volume	mean_pLDDT	PAE_within	verdict	rank
44	0.601	1351.612	83.265484	5.276795	keep	1
4	0.214	206.018	92.453636	1.834711	keep	2
1	0.205	478.850	89.937500	2.631944	keep	3
73	0.053	524.600	84.103077	3.360947	keep	4
7	0.039	707.454	89.257222	2.762346	keep	5

CP: top five pockets by druggability score

pocket	Druggability Score	Volume	mean_pLDDT	PAE_within	verdict	rank
6	0.564	622.200	95.328696	2.173913	keep	1
1	0.463	278.990	97.709375	1.507812	keep	2
32	0.305	522.499	97.574737	1.620499	keep	3
8	0.269	1385.764	95.706333	2.247778	keep	4
3	0.171	653.609	95.828000	1.857500	keep	5

gene	uniprot	module2_rank	n_domains	inter_domain_PAE	pdb_id	rmsd_confident	best_pocket_score	best_pocket_verdict	druggability_call
KCNMA1	Q12791	6	2	22.121182	6V5A	1	0.601	keep	Possible but uncertain
CP	P00450	5	1	NaN	4ENZ	0.62	0.564	keep	Structure supports a small-molecule pocket

To find pockets, I ran fpocket on each AlphaFold model. A score above 0.5 is generally treated as druggable. As per the provided code, each identified pocket was rejected if the mean pLDDT of its lining residues was below 70, considered suspect if the PAE among its lining residues was above 10 Å, and kept otherwise.

For KCNMA1, I made the call "Possible but uncertain". The identified residues (79–364) all sit in the transmembrane domain, which is the half of KCNMA1 that 6V5A does not cover, creating uncertainty around the veracity of the original AlphaFold prediction. Without experimental validation, it is hard to call this pocket ‘real’. For CP, I would say that the "Structure supports a small-molecule pocket". The case for CP is strong from both a prediction and experimental level: The model is highly confident, it matches the crystal at 0.62 Å, and its top pocket (pocket 6, score 0.564) passed both checks with a mean pLDDT of 95.3 and a PAE of 2.2 Å. The only caveat is that the druggability score is just slightly above the 0.5 cutoff.

Ligand site

There was no ligand check for KCNMA1. For CP, the check found two ligands, both copies of NAG (N-acetylglucosamine), at sites A1101 and A1111. These are not covered by the pocket identified in the previous step. The real functional site of CP is its copper centers, which the ligand check skipped because it filters out metal ions. 4ENZ contains six coppers. Repeating the check by hand against the coppers, the top two pockets both sit within 2 Å of a copper position. The best pockets are not where the NAG ligands sit, but they are at an actually structurally important point.

Integration with Modules 2 and 3

The structural results partly agree with the module 2 and the module 3 tractability prior, with caveats. KCNMA1 and CP were ranked 6th and 5th in module 2, respectively. In Module 3, KCNMA1 moved up

to 3rd with a strong small molecule score (p_sm 0.75), and its nearest drugged proteins were other potassium channels. CP moved up to 4th with a weaker small molecule score (p_sm 0.405). After checking with AlphaFold, it seems that CP has the stronger evidence for a real small-molecule pocket, given KCNMA1's best pocket cannot be confirmed with the structures available. However, the biological realities of the proteins are still relevant. As discussed in Module 2, KCNMA1 belongs to the group of smooth muscle genes which may be showing up because normal samples contain more of the muscle layer of the colon wall than tumor samples do. For CP, the prior concern was that it is made mostly by the liver and circulates in the blood, so the CP measured in a tissue sample could come from blood rather than the actual colon cells. I would keep both targets in the pipeline. CP has the most solid structural case, but the biology still has to show that the colon cells themselves are making it before a drug against it makes sense. KCNMA1 is also a potential candidate depending on further experimental evidence.

Limitations

The biggest one is the missing membrane, which affects KCNMA1. KCNMA1 is a membrane protein, but the AlphaFold model is one chain with no membrane. This is the main reason the top pocket cannot be trusted on its own and would need experimental validation. The other big limitation affects CP: it is a copper binding glycoprotein, but AlphaFold does not represent metals (‘Missing pieces’ on the slide). This could be remedied by using a model like Boltz-2.

Anchor check

As per the assignment notebook, we used the same process on p53. The AlphaFold model of p53 matched the 1TUP crystal structure with an RMSD of 0.68 Å over 192 confident residues. Of the hotspot positions, C176, H179, C238 and C242 are zinc ligands. R248 and R273 directly contact DNA. This validates the Module 3 variant scores: positions 176, 238 and 242 were lowly tolerated positions, and R248 was in the bottom 5% of variants, so the language model picked out the zinc site and the DNA-binding surface with no inherent structure at all.

Citations

[1] American Cancer Society. "Colorectal Cancer Statistics." *American Cancer Society*, 14 Jan. 2026, www.cancer.org/cancer/types/colon-rectal-cancer/about/key-statistics.html.

[2] UniProt entries Q12791 (KCNMA1) and P00450 (CP). uniprot.org

[3] Geng Y, et al. Coupling of Ca^{2+} and voltage activation in BK channels through the αB helix/voltage sensor interface. PNAS. 2020;117:14512–14521. DOI: 10.1073/pnas.1908183117

[4] Samygina VR, et al. Ceruloplasmin: macromolecular assemblies with iron-containing acute phase proteins. PLoS One. 2013;8:e67145. DOI: 10.1371/journal.pone.0067145